

```
*****default*****
ben-marshall Uart
links found nanoV
07/11/25
*****default*****
```

```
devel@pi5-90:~ $ git clone https://github.com/develone/uart.git -b dev-test
```

```
devel@pi5-90:~ $ cd uart
```

```
devel@pi5-90:~/uart $ make rx tx
```

```
verilog -o work/sim-rx.bin rtl/uart_rx.v test/tb_rx.v
```

```
test/tb_rx.v:37: vvp.tgt sorry: procedural continuous assignments are not yet fully supported. The
RHS of this assignment will only be evaluated once, at the time the assignment statement is
executed.
```

```
vvp -l work/sim.log work/sim-rx.bin
```

```
VCD info: dumpfile ./work/waves-rx.vcd opened for output.
```

|     |    |                                              |
|-----|----|----------------------------------------------|
| 1/  | 0/ | 1 [PASS] Expected 00100100 and got 00100100  |
| 2/  | 0/ | 2 [PASS] Expected 10000001 and got 10000001  |
| 3/  | 0/ | 3 [PASS] Expected 00001001 and got 00001001  |
| 4/  | 0/ | 4 [PASS] Expected 01100011 and got 01100011  |
| 5/  | 0/ | 5 [PASS] Expected 00001101 and got 00001101  |
| 6/  | 0/ | 6 [PASS] Expected 10001101 and got 10001101  |
| 7/  | 0/ | 7 [PASS] Expected 01100101 and got 01100101  |
| 8/  | 0/ | 8 [PASS] Expected 00010010 and got 00010010  |
| 9/  | 0/ | 9 [PASS] Expected 00000001 and got 00000001  |
| 10/ | 0/ | 10 [PASS] Expected 00001101 and got 00001101 |

```
BIT RATE : 115200b/s
```

```
CLK PERIOD : 20ns
```

```
CYCLES/BIT : 434
```

```
SAMPLE PERIOD : 8680
```

```
BIT PERIOD : 8680ns
```

```
Test Results:
```

```
PASSES: 10
```

```
FAILS : 0
```

```
Finish simulation at time 879040
```

```
iverilog -o work/sim-tx.bin rtl/uart_tx.v test/tb_tx.v
```

```
vvp -l work/sim.log work/sim-tx.bin
```

```
VCD info: dumpfile ./work/waves-tx.vcd opened for output.
```

```
Send data 00100100 at time 40
```

```
Send data 10000001 at time 1041870
```

```
Send data 00001001 at time 2083690
```

```
Send data 01100011 at time 3125510
```

```
Send data 00001101 at time 4167330
```

```
Send data 10001101 at time 5209150
```

```
Send data 01100101 at time 6250970
```

```
Send data 00010010 at time 7292790
```

```
Send data 00000001 at time 8334610
```

```
Send data 00001101 at time 9376430
```

```
Send data 01110110 at time 10418250
```

```

Send data 00111101 at time      11460070
Send data 11101101 at time      12501890
Send data 10001100 at time      13543710
Send data 11111001 at time      14585530
Send data 11000110 at time      15627350
Send data 11000101 at time      16669170
Send data 10101010 at time      17710990
Send data 11100101 at time      18752810
Send data 01110111 at time      19794630
BIT RATE :      9600b/s
BIT PERIOD:    104166ns
CLK PERIOD:     20ns
CYCLES/BIT:     5208
Finish simulation at time      20836450

```

```

devel@pi5-90:~/nanoV/uart $ cd ~/uart/work/
devel@pi5-90:~/uart/work $ ls
sim.log  sim-rx.bin  sim-tx.bin  waves-rx.vcd  waves-tx.vcd

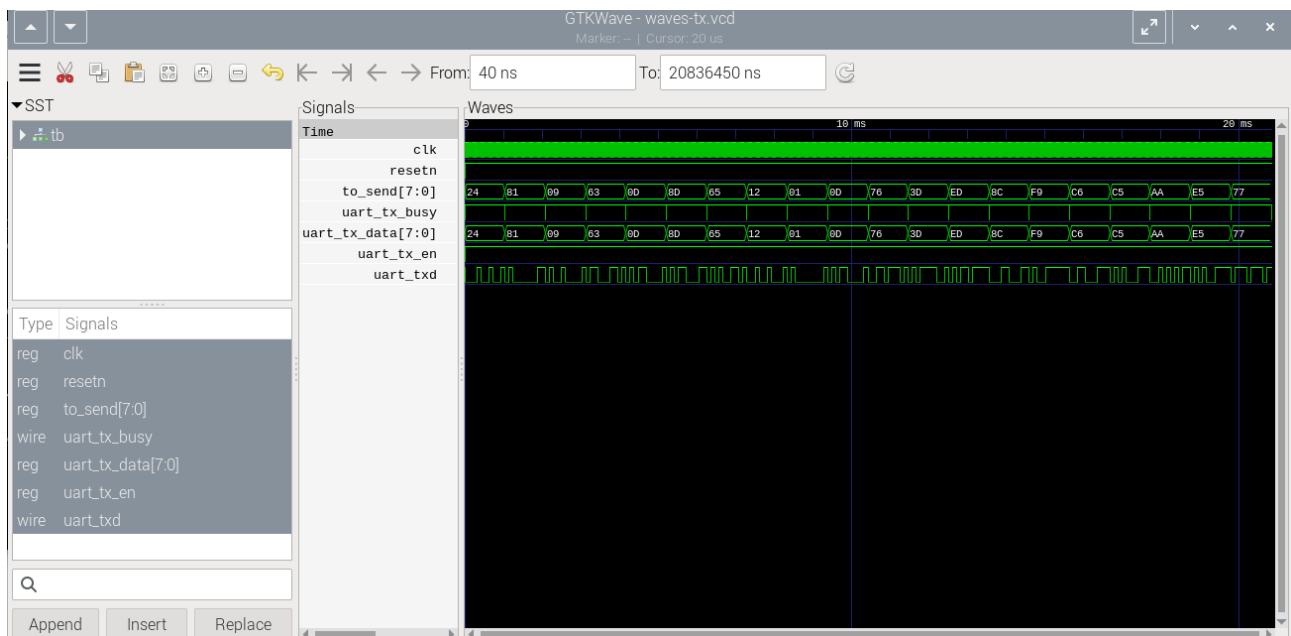
```

```
devel@pi5-90:~/uart/work $ gtkwave waves-tx.vcd
```

GTKWave Analyzer v3.3.118 (w)1999-2023 BSI

[40] start time.

[20836450] end time.

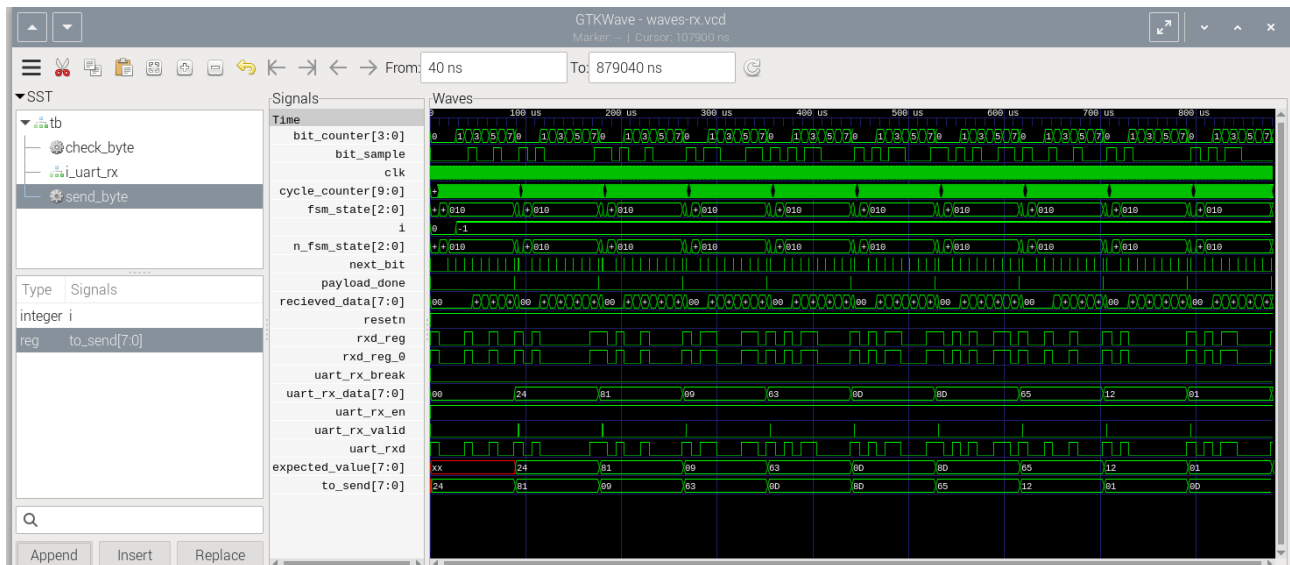


waves-tx-vcd

```
devel@pi5-90:~/uart/work $ gtkwave waves-rx.vcd
```

GTKWave Analyzer v3.3.118 (w)1999-2023 BSI

[40] start time.  
[879040] end time.



waves-rx-vcd